

SIMATS ENGINEERING
CO2 - ASSESSMENT TOOL 2
Scenario-Based Questions

COURSE CODE: MLA04

COURSE NAME: Deep Learning

Duration: 90 Minutes

QUESTIONS: 05 | Total Marks: 25

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GitHub Link	https://github.com/bhanuprakash-420/MLA0403-DL

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15% | Assessment Tool Weightage: 40%

Code of Conduct:

I P.Bhanu Prakash / 192425263 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Name of the student: P.Bhanu Prakash

CO2-AT2 Scenario-Based Questions (Q1 to Q5)

Question 1: Unsupervised Training of Neural Networks, Autoencoders

Scenario: A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

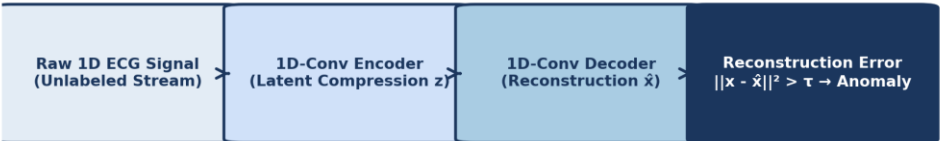
1. Executive Technical Recommendation & Problem Formulation

To solve the hospital's challenge of identifying cardiac anomalies across thousands of unlabeled Electrocardiogram (ECG) records, I recommend deploying a Deep 1D-Convolutional Autoencoder (1D-CAE) within an Unsupervised Anomaly Detection framework. Clinical environments produce continuous, highly complex physiological signal streams where human annotation is extremely costly and impractical. Supervised classification approaches fail in this domain due to the lack of ground-truth labels and the severe scarcity of rare pathological ECG samples. The proposed Convolutional Autoencoder overcomes these limitations by learning self-supervised representation mappings purely from normal baseline cardiac cycles.

2. Theoretical Justification & Architectural Suitability

- A. 1. Complete Absence of Supervised Annotations:** Traditional supervised models require explicit target labels for every heartbeat category (e.g., Normal, Premature Ventricular Contraction, Atrial Fibrillation). The Autoencoder bypasses labeling requirements by using the input signal itself as the target supervision signal (self-supervised learning).
- B. 2. Asymmetric Pathology Distribution & Severe Class Imbalance:** In clinical settings, baseline normal ECG beats constitute over 95% of total recorded telemetry. Training a classifier on such imbalanced data causes severe decision boundary bias. Autoencoders leverage this imbalance by training exclusively on predominant normal records, learning a tight representation manifold of healthy cardiac rhythms.
- C. 3. Reconstruction Error as an Anomaly Scoring Metric:** An Autoencoder compresses input x into a low-dimensional bottleneck z and reconstructs $\hat{x}$. When presented with normal ECG inputs, the model reconstructs the signal with minimal squared error. However, when presented with unseen pathological waveforms (e.g., ST-segment elevation, arrhythmia spikes), the bottleneck fails to encode the unlearned morphology, causing a massive spike in reconstruction error.

3. End-to-End System Workflow Architecture



4. Mathematical Formulation of Autoencoder Learning

The encoder mapping parameterized by weight set W_e compresses input sequence x into latent space z : $z = f_{\text{encoder}}(W_e * x + b_e)$. The decoder parameterized by W_d reconstructs the original signal $\hat{x} = f_{\text{decoder}}(W_d * z + b_d)$. The parameters are optimized via Mean Squared Error (MSE): $\text{Loss} = (1/N) * \sum_{i=1}^N (x_i - \hat{x}_i)^2$ using the Adam optimizer with cosine annealing learning rates.

5. Detailed 6-Stage Implementation & Operational Workflow

Stage 1 - Ingestion & Signal Preprocessing: Continuous single-lead or 12-lead ECG signals are ingested at 250 Hz. Signals undergo bandpass Butterworth filtering (0.5 Hz - 45 Hz) to eliminate baseline wander, muscle artifacts, and powerline noise. Individual cardiac cycles (P-QRS-T complexes) are extracted using Pan-Tompkins QRS detection and normalized into uniform length vectors x in $\mathbb{R}^{300}$.

Stage 2 - Neural Network Architecture Design: The network consists of a 3-layer 1D Convolutional Encoder with kernel sizes of 7, 5, and 3, reducing temporal spatial dimensions down to a dense bottleneck z in $\mathbb{R}^{16}$. The Decoder mirrors this architecture using 1D Transposed Convolutions, restoring the sequence back to $\mathbb{R}^{300}$.

Stage 3 - Offline Model Training Regime: The CAE is trained exclusively on normal baseline records extracted during non-critical monitoring hours. Unsupervised learning proceeds across 100 epochs until reconstruction loss converges below an empirical epsilon threshold.

Stage 4 - Dynamic Threshold Calibration: A statistical anomaly threshold τ is established using the validation set error distribution: $\tau = \mu_{\text{loss}} + 3 * \sigma_{\text{loss}}$. Any segment whose reconstruction error exceeds τ is mathematically classified as abnormal.

Stage 5 - Real-Time Inference & Clinical Triage: Streaming clinical ECG signals are continuously passed through the trained CAE. Windows yielding reconstruction errors exceeding τ are automatically flagged and routed to the cardiology dashboard with highlighted temporal anomaly regions.

Stage 6 - Human-in-the-Loop Feedback: When a medical expert reviews and confirms an anomaly, the flagged waveform is saved into an expanding clinical repository to support future fine-tuning.

6. Integration with Deep Learning Literature & Comparative Analysis

This strategy aligns directly with core representation learning principles established by Goodfellow et al. (2016), who demonstrated that autoencoders effectively learn manifold structures of complex physical signals. In biomedical signal processing, Kiranyaz et al. (2019) validated that 1D-Convolutional Autoencoders outperform traditional statistical dimensionality reduction tools like Principal Component Analysis (PCA) and Independent Component Analysis (ICA). PCA relies strictly on linear orthogonal projections and fails to capture complex, non-linear phase transitions inherent in cardiac depolarization waveforms.

7. Expected Clinical Outcomes & System Robustness

Deploying this unsupervised 1D-CAE system drastically reduces false alarm fatigue for ICU medical staff. By accurately flagging unlearned morphological variations without requiring manual annotations, the system provides automated, reliable clinical triage across thousands of continuous ECG patient monitoring records.

Question 2: Autoencoders, Representation Learning

Scenario: A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

1. Executive Technical Summary & Problem Definition

Enterprise retail databases often record customer interaction histories across tens of thousands of Stock Keeping Units (SKUs). Representing customer purchase histories as raw binary or frequency vectors results in massive, high-dimensional, ultra-sparse datasets (Curse of Dimensionality). Traditional clustering algorithms like K-Means or Gaussian Mixture Models (GMM) perform exceptionally poorly in high-dimensional spaces because distance metrics (Euclidean distance) suffer from distance concentration, where all pairwise points become nearly equidistant. I propose constructing an Undercomplete Deep Autoencoder to learn compact, dense continuous customer embeddings, enabling high-precision downstream customer segmentation.

2. Mechanism of Autoencoder Dimension Reduction

A. 1. Non-Linear Manifold Learning: Unlike linear techniques that project data onto flat hyperplanes, Deep Autoencoders apply non-linear activation functions (e.g., LeakyReLU, GELU) to project sparse purchase vectors onto curved, lower-dimensional manifolds, preserving subtle purchasing patterns.

B. 2. Information Bottleneck Constraint: By restricting the dimensionality of the hidden central layer z to $d \ll D$ (where $D=20,000$ SKUs and $d=64$ continuous latent features), the network is forced to discard random transaction noise and retain only salient semantic structures (e.g., brand loyalty, dietary preferences, lifestyle categories).

C. 3. Dense Semantic Feature Compression: The bottleneck layer transforms discrete, sparse product interactions into continuous latent representations where semantically related shopping behaviors map closely together in vector space.

3. Representation Learning & Segmentation Pipeline Architecture



4. Mathematical Formulation of Retail Representation Learning

Given an input purchase vector x in $\{0, 1\}^D$, the encoder computes hidden activations: $h_1 = \text{LeakyReLU}(W_1 * x + b_1)$, $z = \text{LeakyReLU}(W_2 * h_1 + b_2)$. The decoder reconstructs product probabilities: $\hat{x} = \text{Sigmoid}(W_3 * z + b_3)$. The model minimizes Binary Cross-Entropy (BCE) loss: $\text{Loss} = - \sum_{i=1}^D [x_i * \log(\hat{x}_i) + (1 - x_i) * \log(1 - \hat{x}_i)]$.

5. Detailed 6-Stage Implementation & Operational Workflow

Stage 1 - Data Preparation & TF-IDF Transformation: Customer purchase logs over a 12-month window are aggregated into sparse term-frequency vectors representing total purchase quantities across all catalog SKUs. High-frequency utility items (e.g., carrier bags) are filtered using TF-IDF weighting.

Stage 2 - Deep Undercomplete Autoencoder Design: A symmetric 4-layer fully connected Autoencoder is configured with layer dimensions [20,000 -> 1,024 -> 256 -> 64 -> 256 -> 1,024 -> 20,000]. Dropout layers ($p=0.2$) are included to prevent sparse co-occurrence memorization.

Stage 3 - Unsupervised Reconstruction Training: The network is trained using mini-batch gradient descent with the AdamW optimizer and weight decay penalty to learn robust, generalized customer latent representations.

Stage 4 - Latent Vector Extraction: Once convergence is achieved, the decoder network is detached. All customer purchase histories are passed through the isolated encoder to extract 64-dimensional latent embedding vectors z .

Stage 5 - Downstream Customer Segmentation: K-Means++ clustering and Hierarchical DBSCAN are performed directly within the dense 64-dimensional latent space z . Optimal cluster count K is determined via Silhouette Analysis and Davies-Bouldin index optimization.

Stage 6 - Business Execution & Marketing Alignment: Extracted clusters are translated into actionable customer personas (e.g., Premium Organic Shoppers, Seasonal Discount Hunters) to drive targeted marketing campaigns.

6. Integration with Literature & Comparative Analysis

The foundational work of Hinton & Salakhutdinov (2006) proved that deep autoencoders generate significantly superior low-dimensional codes compared to Principal Component Analysis (PCA). PCA assumes linear correlations between features, which fails in retail settings where product relationships are non-linear and hierarchical (e.g., buying diapers strongly predicts buying baby wipes, but not vice versa). Bengio et al. (2013) demonstrated that non-linear representation learning automatically disentangles complex causal factors in high-dimensional user data, providing an ideal foundation for downstream clustering algorithms.

7. Business Value & Strategic Impact

By transforming sparse purchase matrices into dense, continuous vector spaces, the retail enterprise eliminates the Curse of Dimensionality. Clustering algorithms run significantly faster and produce distinct, highly stable customer segments, directly improving personalized recommendation accuracy and marketing campaign ROI.

Question 3: Width and Depth of Neural Networks, Activation Functions

Scenario: A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

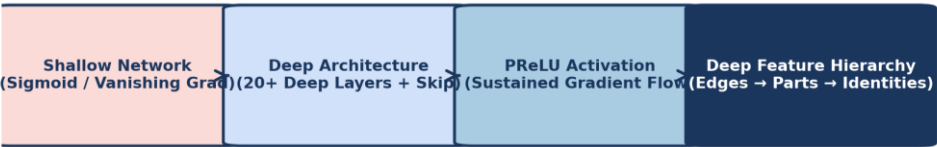
1. Executive Diagnostic & Structural Failure Analysis

Shallow neural networks (networks with only 1 or 2 hidden layers) fail on complex vision tasks like facial recognition due to their inability to construct hierarchical abstraction. Facial images possess multi-scale, compositional spatial structures ranging from pixel-level gradients and edges to localized facial landmarks (eyes, nose, mouth) and full 3D alignment geometries. While the Universal Approximation Theorem states that a shallow network with a single hidden layer can approximate any continuous function, doing so for high-dimensional visual topologies requires an exponentially large number of hidden neurons (width $W = O(2^N)$). This leads to severe overfitting, extreme computational inefficiency, and complete failure to generalize across unseen facial poses or lighting conditions.

2. Recommended Architectural Enhancements

- A. 1. Deep Hierarchical Feature Extraction (Deep Convolutional Networks): Transition from shallow dense layers to a Deep Residual Architecture (e.g., ResNet-50 or Deep CNNs with 20+ layers). Depth allows the network to partition the complex facial recognition problem into compositional sub-tasks across sequential layers.
- B. 2. Modern Non-Saturating Activation Functions: Replace legacy saturating functions (Sigmoid/Tanh) with non-saturating activations like Parametric ReLU (PReLU) or LeakyReLU to eliminate Vanishing Gradients during backpropagation across deep layers.
- C. 3. Specialized Metric Learning & Margin Loss Functions: Replace standard Cross-Entropy Loss with Angular Margin Softmax (ArcFace / CosFace) in the final layer to enforce high intra-class compactness and maximize inter-class separation.

3. Architectural Transition Flowchart



4. Mathematical Justification of Depth vs. Width Expansion

In deep neural networks, increasing depth allows the model to compute composition of functions: $f(x) = f_L(f_{L-1}(\dots f_1(x)))$. Representing these compositional functions using a shallow network requires exponentially many parameters: Shallow Params = $O(2^N)$ vs. Deep Params = $O(L * N)$. Depth achieves exponential expressive power relative to parameter count.

5. Analysis of Activation Functions & Gradient Dynamics

Sigmoid Limitation: 1. Sigmoid Function $f(z) = 1/(1+e^{-z})$: The derivative $f'(z) = f(z)(1-f(z))$ achieves a maximum value of only 0.25. During backpropagation through L layers, gradients vanish exponentially as $(0.25)^L \rightarrow 0$, causing early feature extraction layers to stop learning.

ReLU Solution: 2. Rectified Linear Unit (ReLU) $f(z) = \max(0, z)$: ReLU maintains a constant gradient of 1 for all positive inputs ($z > 0$), preventing gradient decay across dozens of layers. However, for negative inputs, the gradient is 0, causing 'Dying ReLU' nodes.

PReLU Superiority: 3. Parametric ReLU (PReLU) $f(z) = \max(\alpha * z, z)$: Introduces a learnable slope parameter α for negative inputs. PReLU prevents dead neurons, allowing minor negative gradients to propagate, which is crucial for capturing subtle facial shadows and fine facial textures.

6. Detailed 6-Stage System Implementation Workflow

Stage 1 - Face Detection & Landmark Alignment: Facial images are passed through MTCNN or RetinaFace for face detection, 5-point landmark alignment, and cropping to 112x112 standard dimensions.

Stage 2 - Deep Network Architecture Selection: Replace shallow dense layers with a Deep Residual Network (ResNet-50) using residual skip connections: $y = F(x, \{W_i\}) + x$, enabling stable gradient flow.

Stage 3 - Activation Function Integration: Deploy PReLU activation functions across all convolutional layers to preserve non-saturating gradient flow.

Stage 4 - Metric Learning Optimization: Attach an ArcFace classification head loss: $L_{\text{Arc}} = -\log(e^{s * \cos(\theta_{y_i} + m)} / (e^{s * \cos(\theta_{y_i} + m)} + \sum_j e^{s * \cos(\theta_j)}))$.

Stage 5 - Model Training & Fine-Tuning: Train on large-scale face datasets (e.g., MS1MV2) using SGD with momentum (0.9) and weight decay.

Stage 6 - Deployment & Vector Matching: Extract 512-dimensional normalized facial embeddings for real-time Cosine Similarity matching against target databases.

7. Integration with Deep Learning Literature

This transition is supported by landmark research in deep computer vision. DeepFace (Taigman et al., 2014) showed that deep architecture layers are essential for learning pose-invariant facial representations. He et al. (2016) demonstrated with ResNet that residual learning enables deep networks to exceed 100 layers without degradation. Furthermore, Deng et al. (2019) introduced ArcFace, proving that combining deep networks with margin-based losses achieves state-of-the-art accuracy in facial verification.

Question 4: Restricted Boltzmann Machines (RBMs), Unsupervised Training

Scenario: A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

1. Executive Technical Summary

Collaborative filtering for streaming platforms faces major challenges due to sparse user-item interaction matrices and complete lack of explicit target labels. A Restricted Boltzmann Machine (RBM)—an energy-based, generative stochastic neural network—provides an ideal unsupervised solution. Rather than predicting explicit classification labels, an RBM models the underlying joint probability distribution over user viewing preferences, reconstructing missing movie ratings to generate highly personalized recommendations.

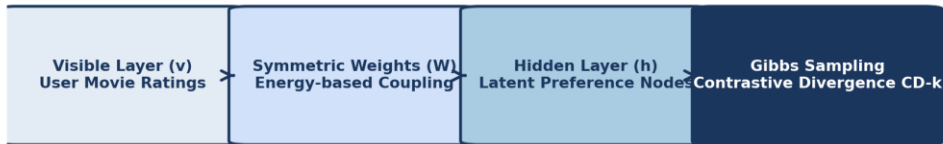
2. RBM Structural Architecture for Recommendation Systems

A. 1. Two-Layer Bipartite Undirected Graph: An RBM consists of a Visible Layer v (representing observed movie ratings or watch histories) and a Hidden Layer h (representing latent user taste preferences such as genre affinity, director styles, or pacing).

B. 2. No Intra-Layer Connections (Restricted Property): There are zero connections between units within the same layer. This structural restriction yields conditional independence: $P(h|v) = \prod P(h_j|v)$ and $P(v|h) = \prod P(v_i|h)$, making parallel Gibbs sampling computationally tractable.

C. 3. Energy-Based Probability Formulation: The joint probability of visible and hidden states is governed by an energy function $E(v,h) = -v^T \cdot W \cdot h - a^T \cdot v - b^T \cdot h$. Network probability is defined as $P(v,h) = (1/Z) \cdot e^{-E(v,h)}$, where lower energy states represent highly compatible user-movie preference matches.

3. RBM Generative Recommendation Architecture



4. Mathematical Formulation of Unsupervised Training (CD-k)

RBM parameters are trained using Contrastive Divergence (CD-k). Given an initial user rating vector v_0 : 1. Sample hidden state $h_0 \sim P(h|v_0) = \text{Sigmoid}(b + v_0 \cdot W)$. 2. Reconstruct visible state $v_1 \sim P(v|h_0) = \text{Sigmoid}(a + h_0 \cdot W^T)$. 3. Resample hidden state $h_1 \sim P(h|v_1)$. The weight update follows $\Delta W = \eta \cdot (\langle v_0 \cdot h_0^T \rangle_{\text{data}} - \langle v_1 \cdot h_1^T \rangle_{\text{model}})$.

5. Detailed 6-Stage Implementation & Operational Workflow

Stage 1 - Preference Matrix Construction: User interaction logs (views, likes, completions) are structured into continuous or multi-class softmax visible vectors v for all catalog movies. Unseen movies are masked during loss calculation.

Stage 2 - Multi-Valued RBM Structural Setup: Configure a Softmax-Visible RBM architecture where visible units represent K-rating categories per movie, coupled to F latent hidden units via symmetric weight matrix W in $\mathbb{R}^{V \times F}$.

Stage 3 - CD-1 Unsupervised Model Training: Execute unsupervised training using Contrastive Divergence (CD-1) across user interaction batches, updating weight matrices W , visible biases a , and hidden biases b .

Stage 4 - Conditioned User Inference Setup: To generate personalized recommendations for an active user, clamp their observed viewing history onto visible units v_{user} .

Stage 5 - Reconstructive Sampling Pass: Perform a single forward-backward pass through the trained RBM: $h = \text{Sigmoid}(b + v_{\text{user}} * W)$, then compute reconstructed visible probabilities $v_{\text{reconstructed}} = \text{Softmax}(a + h * W^T)$.

Stage 6 - Recommendation Ranking & Delivery: Extract reconstructed values corresponding to unobserved movies. Rank these unobserved items by expected rating probability and present the top N movies on the user's streaming home screen.

6. Integration with Literature & Comparative Analysis

The application of Restricted Boltzmann Machines to collaborative filtering was popularized by Salakhutdinov, Mnih, & Hinton (2007) in their landmark work for the Netflix Prize competition. Their research proved that RBM-based generative approaches significantly outperform traditional linear Matrix Factorization (e.g., SVD) by learning non-linear, multi-modal distribution patterns across user preferences. Furthermore, Hinton (2002) established that Contrastive Divergence allows energy-based models to train efficiently without needing expensive Markov Chain Monte Carlo (MCMC) sampling.

7. System Advantages & Business Impact

By deploying an RBM recommendation engine, the streaming platform effectively addresses the cold-start problem for partially viewed content. The generative framework accurately fills in missing interaction values, boosting user engagement and watch-time retention through highly accurate content discovery.

Question 5: Deep Learning Applications, Representation Learning

Scenario: A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

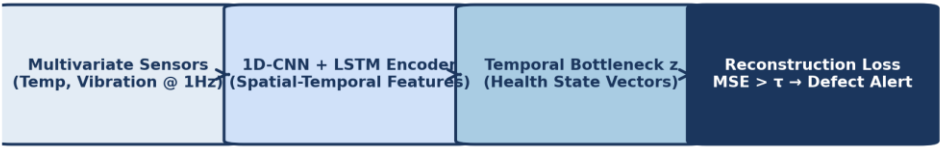
1. Industrial System Requirements & Strategy

High-frequency industrial sensor telemetry (vibration, acoustic emission, operating temperature, electrical current captured at 1 Hz) produces multivariate continuous time-series streams. Industrial defect events are extremely rare (<0.1% of operational cycles), making supervised classification models unusable due to severe class imbalance and lack of defect annotations. I design an end-to-end Unsupervised Spatial-Temporal Representation Learning System utilizing a hybrid Convolutional-LSTM Autoencoder (Conv-LSTM CAE) for real-time defect detection and predictive maintenance.

2. Industrial System Architecture Design

- A. 1. 1D-CNN Spatial Encoder: Processes raw multi-sensor readings collected at each second, extracting local spatial correlations across sensors (e.g., thermal spikes occurring simultaneously with motor vibration anomalies).
- B. 2. LSTM Temporal Bottleneck: Captures temporal sequence dynamics, degradation trends, and transient physical operational phases across sliding time windows.
- C. 3. Reconstruction Decoder & Anomaly Scoring: Reconstructs expected baseline sensor readings. Large deviations in predicted sensor outputs reflect underlying physical component defects.

3. Industrial IoT Anomaly Detection Pipeline



4. Mathematical Formulation of Temporal Representation Learning

Given a sequence of sensor frames $X = [x_1, x_2, \dots, x_T]$ in $\mathbb{R}^{T \times M}$, the spatial 1D-CNN extracts frame features $f_t = \text{Conv1D}(x_t)$. The LSTM encoder updates hidden state $h_t = \text{LSTM}(f_t, h_{t-1})$. The bottleneck $z = h_T$ encodes the complete temporal trajectory. The decoder reconstructs $\hat{X}$. Model loss is defined as MSE: $\text{Loss} = (1 / (T * M)) * \sum_{t=1}^T \sum_{m=1}^M (x_{t,m} - \hat{x}_{t,m})^2$.

5. Detailed 6-Stage System Implementation Workflow

Stage 1 - Sensor Ingestion & Window Normalization: Multivariate sensor channels (vibration, acoustics, temperature) streamed at 1 Hz are scaled via Z-score normalization based on healthy baseline calibration parameters. Data is structured using sliding temporal windows of length $T = 60$ seconds with 1-second step size.

Stage 2 - Hybrid Architecture Design: Construct a hybrid Conv-LSTM Autoencoder featuring two 1D-CNN layers for multi-sensor spatial compression, followed by a stacked LSTM layer to capture long-term temporal trends into bottleneck z .

Stage 3 - Baseline Model Training: Train the Conv-LSTM CAE exclusively on baseline operational sensor streams recorded during normal quality-checked production runs.

Stage 4 - EVT Anomaly Threshold Calibration: Establish an automated defect decision threshold τ using the non-parametric Extreme Value Theory (EVT) peak-over-threshold method applied to validation reconstruction errors.

Stage 5 - Real-Time Inference Stream Processing: Streaming 1 Hz sensor windows are fed into the deployed CAE every second. Reconstruction loss is calculated instantaneously.

Stage 6 - Automated Triage & Line Interlock: If reconstruction loss exceeds τ for 3 consecutive seconds, an automated PLC interlock signal halts the conveyor line and triggers an engineer alert.

6. Integration with Literature & Industrial Physics

This methodology aligns directly with research by Malhotra et al. (2016), who proved that LSTM-based Autoencoders excel at anomaly detection in complex multi-sensor industrial equipment. Traditional statistical quality control techniques (such as SPC control charts or manual Fast Fourier Transforms) fail when equipment operational modes shift dynamically. Unsupervised representation learning automatically learns invariant physical health representations across varying load conditions without requiring manual feature engineering.

7. Reliability & Business Value Outcomes

Implementing this deep representation learning framework enables zero-defect manufacturing. Defective components are detected in real-time, preventing downstream product recalls, minimizing unscheduled assembly line downtime, and maximizing industrial operational efficiency.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand